

CS & IT ENGINEERING



Computer Network

IPv4 Header

Lecture No. - 07

By - Abhishek Sir





Recap of Previous Lecture



Topic

Time-to-Live

Topic

Protocol Type

Topic

Checksum



Topics to be Covered



Topic

IPv4 Options

Topic

ICMP

ABOUT ME



Hello, I'm **Abhishek**

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- 12 years of GATE CS teaching experience

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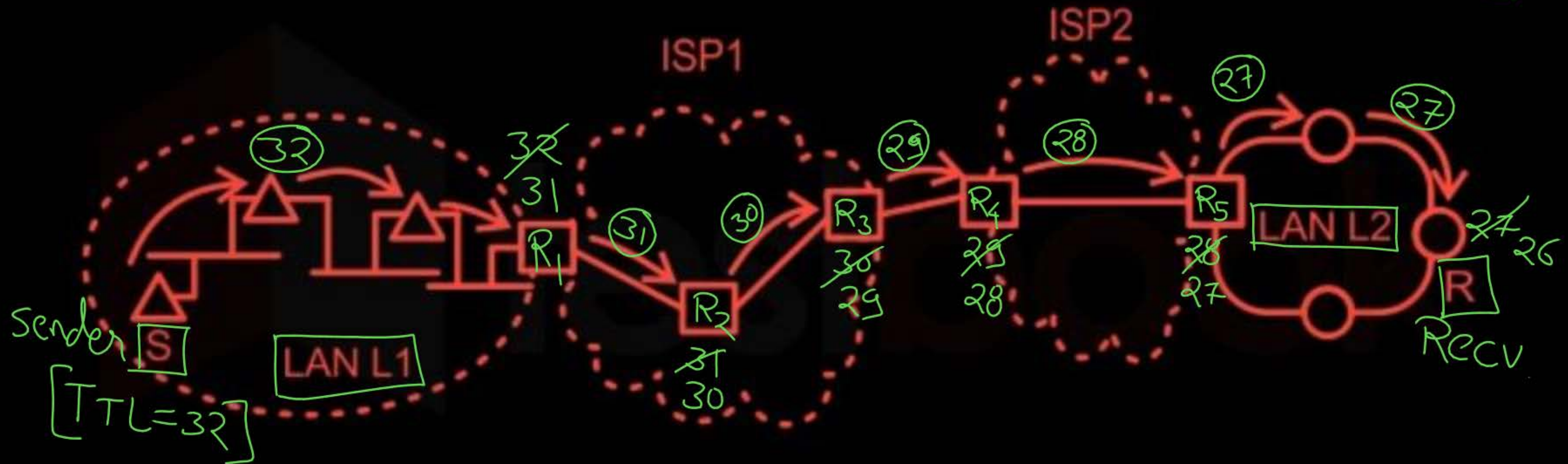
[NAT]

IIT-KGP [GATE-2014][1 Mark]



#Q. In the diagram shown below L1 is an Ethernet LAN and L2 is a Token-Ring LAN. An IP packet originates from sender S and traverses to R, as shown. The link within each ISP, and across two ISPs, are all point to point optical links. The initial value of TTL is 32. The maximum possible value of TTL field when R receives the datagram is ____.

IIT-KGP Ans = 26





Topic : Header Checksum



→ Checksum verification is performed at every intermediate router and destination host

→ if error is detected in IPv4 header
(computed checksum is nonzero)
then discards the datagram and may send ICMP error to source host

→ After processing the IPv4 datagram,
New Checksum is generated by every intermediate router, before forwarding it.



Topic : IPv4 Header



→ At intermediate Router: (for every received IPv4 datagram)

1. Verify Header Checksum,
if error detected, discard the datagram
and may send ICMP error to source host

2. $TTL = TTL - 1$; [Decrement TTL by one]
if $TTL \leq 0$, then discard the datagram
and send ICMP error to source host $\Rightarrow$ Time Exceeded

3. Process the datagram,
to determine next forwarding interface

4. Update Checksum field with new generated checksum
and forward the datagram



Topic : IPv4 Header



→ At destination host : (for every received IPv4 datagram)

1. Verify Header Checksum,
if error detected, discard the datagram
and may send ICMP error to source host

2. $TTL = TTL - 1$; [Decrement TTL by one]
if $TTL < 0$, then discard the datagram
and send ICMP error to source host

⇒ optional for
Dest. Host

3. Process the datagram,
to determine higher level protocol

4. Forward the payload to higher level protocol



Topic : Source IP Address



- Source IP Address is 32-bit field in IPv4 Header
- Fourth word of IPv4 Header [Word size = 4 byte]
- Assigned by source host only
- Does not change during routing
- May update in case of NAT device



Topic : Destination IP Address

- Destination IP Address is 32-bit field in IPv4 Header
- Fifth word of IPv4 Header [Word size = 4 byte]
- Assigned by source host only
- Does not change during routing
- May update in case of NAT device

[MSQ]

11SC

[GATE-2024][1 Mark]



#Q. Which of the following fields of an IP header is/are always modified by any router before it forwards the IP packet?

- ☒ A Source IP Address
- ☒ B Protocol Type
- ☐ C Time to Live (TTL)
- ☐ D Header Checksum

Ans: C & D

[MSQ]

[GATE-2024][1 Mark]



#Q. Which of the following fields is/are modified in the IP header of a packet going out of a network address translation (NAT) device from an internal network to an external network?

→ Public N/w (Internet) | Private N/w

- ☒ A Header Checksum
- ☒ B Source IP (from internal N/w to External N/w)
- ☐ C Destination IP (from external N/w to internal N/w)
- ☐ D Total Length

Ans: A & B

[MCQ]

[GATE-2015][1 Mark]



#Q. Which one of the following fields of an IP header is NOT modified by a typical IP router ?

☒ A Checksum

☐ B Source address

☒ C Time to Live (TTL)

☒ D Length

Ans: B

[MCQ]

[GATE-2014][1 Mark]



#Q. Host A (on TCP/IPv4 network A) sends an IP datagram D to host B (also on TCP/IPv4 network B). Assume that no error occurred during the transmission of D. When D reaches B, which of the following IP header field(s) may be different from that of the original datagram D?

(i) TTL

(ii) Checksum

(iii) Fragment Offset

A (i) only

B (i) and (ii) only [In case of No fragmentation]

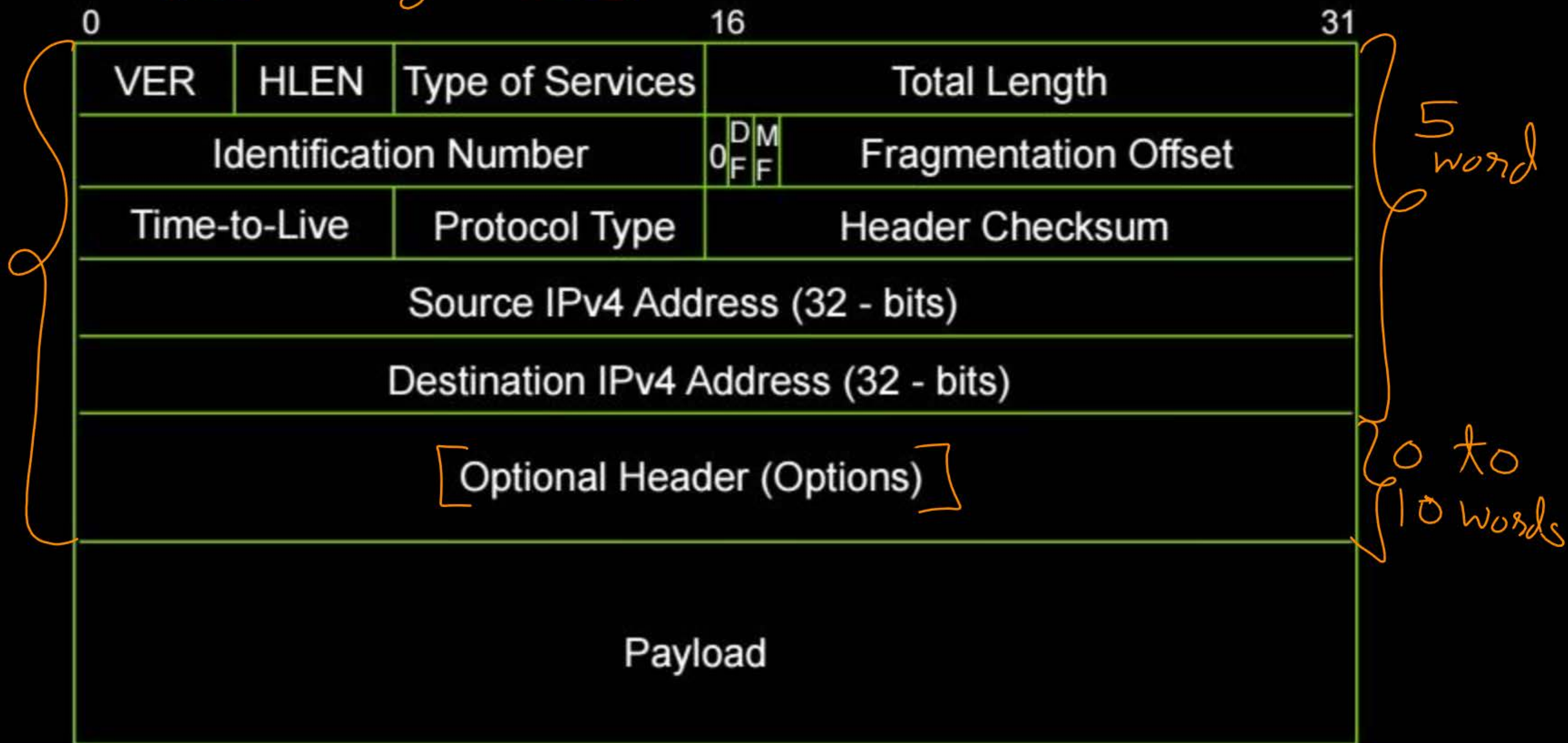
C (ii) and (iii) only

D (i), (ii) and (iii)

IIT KGP
Ans: D



Topic : IPv4 Packet Header





Topic : IPv4 Optional Header

- Also known as "Options"
- Options are variable in size
- Maximum size of IPv4 optional header is 10 words
[Words of 32 bits, i.e. 40 bytes]
- if options size are not in words
than it uses padding [End-of-Option Option]



Topic : IPv4 Optional Header



→ Options field component :-

1. Option Type (1 byte)

2. Option Length (1 byte)

3. Option Data

→ if (HLEN > 5)
→ Compulsary

} → optional field



Topic : IPv4 Optional Header

VER | HLEN

BASE Header
[5 Words]

Option Type

Options

Payload



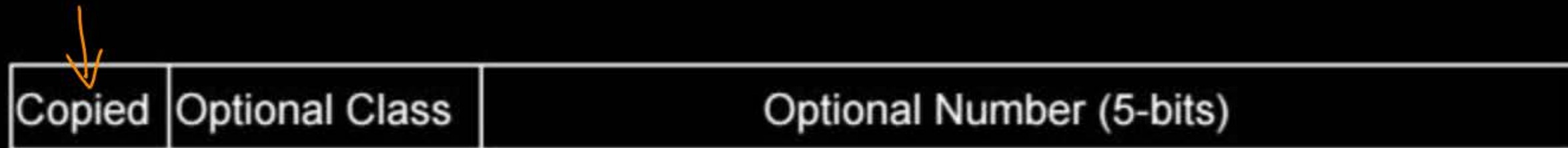
Topic : Option Type



→ Option type field are of size 8-bits (one byte)

→ Option type field are divided into three sub-fields

1. Copied (1-bit)
2. Option Class (2-bits)
3. Option Number (5-bits)



Copied (0) : 0 to 127

Copied (1) : ~~0 to 127~~ 128 to 255



Topic : Option Type



Number	Option Type
<u>0</u>	End of Option [<u>Padding</u>]
1	[No Operation] ✓
7	[Record Route]
68	[Timestamp]
82	[Traceroute]
131	Loose Source Route
137	Strict Source Route



Topic : Option Length



- Option length field are of size 8-bits
- Define total length of Options (in bytes)
[including option type field and option length field itself]



Topic : Record Route



- Used to record the IP address of Internet Routers
[Routers those handle the datagram along the path]
- It can list up to "nine" router IP address



Topic : Record Route



39 Bytes
↑

Type = 1 00000001	Type = 7 00000111	Total Length = 39 00100111	Pointer = 4 00000100
First IP Address (Empty when started)			
Second IP Address (Empty when started)			
Third IP Address (Empty when started)			
Last IP Address (Empty when started)			

Options
(40 byte)

[NAT]

[GATE-2017][1 Mark]



#Q. The maximum number of IPv4 router addresses that can be listed in the record route (RR) option field of an IPv4 header is _____.

Ans = 9

$$\left\lfloor \frac{37}{4} \right\rfloor = 9$$



Topic : Timestamp



→ Similar to the Record Route

→ Every router need to record the IP address and Timestamp

→ 32-bits Timestamp in milliseconds from UTC.

(32 bit)

(32 bit)



$$\left\lfloor \frac{37}{8} \right\rfloor = 4 \text{ Routers}$$



2 mins Summary



Topic

IPv4 Options

Topic

~~ICMP~~

Source Routing
ICMP
DHCP



THANK - YOU

